

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) who has been building exactly what PicnicHealth needs: production healthcare RAG systems, agentic LLM automation pipelines, and HIPAA-conscious data governance - in a clinical AI context. Proven metrics: ~35% contextual retrieval precision improvement, >30% irrelevant retrieval reduction, >90% retrieval-grounded response alignment, and >30% hallucination reduction. Experienced bridging AI engineering and non-technical clinical teams, enabling AI-powered automation of manual operational workflows. Remote-ready and passionate about applying AI to accelerate clinical research.

CORE COMPETENCIES

Healthcare RAG Systems • Agentic LLM Workflows • HIPAA-Conscious AI Governance • Clinical AI Automation • Python & FastAPI • LangChain & LlamaIndex • Predictive ML (scikit-learn, XGBoost) • Docker & Production Packaging • Non-Technical User Enablement • React / TypeScript

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting, SaaS Healthcare | USA

- Built **RAG systems for clinical knowledge retrieval** and healthcare documentation search; improved contextual retrieval precision ~35%, reduced irrelevant retrieval >30%, and achieved >90% retrieval-grounded response alignment - directly applicable to PicnicHealth's clinical research data retrieval challenges.
- Developed **agentic LLM workflows** for multi-step healthcare automation (eligibility checks, care workflow navigation, documentation clarification); improved agent response stability ~25% and reduced hallucinations >30%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient risk scoring; improved recall 15-20% on high-risk categories while holding precision >90%.
- Built and deployed **AI tools for non-technical clinical staff** - Flask API with lightweight web interface for document Q&A; enabled non-engineering teams to adopt AI-assisted search workflows.
- Maintained **HIPAA-conscious data governance**: de-identification, data lineage documentation, evaluation audit trails, and stakeholder-facing system-limitation docs.
- Packaged ML/LLM inference as FastAPI/Flask RESTful services with Docker; reduced post-deployment defects ~30%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech, Video Analytics, Applied NLP | Hyderabad, India

- Built transformer-based video summarization system for 5,000+ sessions; reduced manual review time **60-70%** and achieved ~85% alignment with human-curated highlights - example of automating high-volume manual operational work with AI.
- Deployed document Q&A with RAG for non-technical staff; reduced hallucinated outputs ~30% and raised contextual accuracy >85%.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Enterprise ERP | Hyderabad, India

- Optimized SQL/T-SQL stored procedures; reduced query execution time ~20% and data retrieval latency ~25%.
- Built CI/CD pipelines with Jenkins; reduced deployment errors >30%.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with schema-validated JSON contracts between agents. RAG-grounded CPT/ICD validation against vector-indexed policy documents; audit-ready reasoning traces - directly applicable to clinical study data validation.

Python, LangChain, LlamaIndex, FastAPI, Docker, RAG, multi-agent orchestration

Patient Records Management Web Application

Bottle + SQLite relational healthcare CRUD app with normalized patients/diagnoses schema, foreign-key constraints, and LEFT JOIN + GROUP_CONCAT aggregation for single-query patient+diagnosis retrieval. Demonstrates healthcare data modeling expertise.

Python, SQLite, Bottle, SQL, healthcare data

Manga Lens - Chrome Extension for Real-Time AI Translation shipped

Independently built and shipped full-stack browser extension integrating 4 AI vision providers; 7-day cache, 29-site support. Demonstrates high-agency independent product delivery.

TypeScript, Manifest V3, Claude Sonnet, GPT-4o mini

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, SQL

LLM & GenAI: LangChain, LlamaIndex, RAG, agentic workflows, prompt engineering, evaluation pipelines, guardrails, embeddings, vector search

ML: scikit-learn, XGBoost, PyTorch, HuggingFace Transformers

Infra: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, observability/logging

Domains: Healthcare AI (HIPAA-conscious), Clinical AI Automation, Workflow Automation, AI for Non-Technical Teams